Nelson–Siegel–Svensson Estimation of Greek and Italian Sovereign Discount Curves

Soumadeep Ghosh

Abstract

This paper presents a reproducible comparison of Greek and Italian sovereign discount curves. Official benchmark yields are transformed according to an explicit modelling convention, fitted separately with the Nelson–Siegel–Svensson model, and converted into model-implied discount factors. The empirical tables and figure are generated automatically by the repository pipeline.

1 Discount Factors

Definition 1.1 (Discount factor)**.** *For maturity m, the discount factor $D(m)$ is the present value of one unit payable at maturity under a specified pricing convention.*

Under continuous compounding,

$$D(m) = \exp[-my(m)], \tag{1}$$

where $y(m)$ is the continuously compounded spot yield.

Observed sovereign benchmark yields-to-maturity are not themselves directly observed zero-coupon spot rates. Accordingly, the repository records its transformation convention explicitly and interprets all resulting discount factors as model-implied quantities.

2 Nelson–Siegel–Svensson Specification

Let $\theta = (\beta_0, \beta_1, \beta_2, \beta_3, \tau_1, \tau_2)$ with $\tau_1, \tau_2 > 0$. The NSS spot curve is

$$\begin{aligned} y(m;\theta) = \beta_0 + \beta_1 \frac{1 - e^{-m/\tau_1}}{m/\tau_1} \qquad (2) \\ + \beta_2 \left(\frac{1 - e^{-m/\tau_1}}{m/\tau_1} - e^{-m/\tau_1} \right) \qquad (3) \\ + \beta_3 \left(\frac{1 - e^{-m/\tau_2}}{m/\tau_2} - e^{-m/\tau_2} \right). \qquad (4) \end{aligned}$$

For country $c \in \{\mathrm{GR}, \mathrm{IT}\}$, the model-implied discount curve is

$$\hat{D}_c(m) = \exp[-my(m;\hat{\theta}_c)]. \tag{5}$$

The bilateral comparison is

$$\Delta D(m) = \hat{D}_{\mathrm{GR}}(m) - \hat{D}_{\mathrm{IT}}(m). \tag{6}$$

3 Reproducible Empirical Pipeline

The workflow executes the sequence

$$\text{raw data} \to \text{estimation} \to \text{plot generation} \to \text{tests} \to \text{generated results} \to \text{PDF compilation.}$$

The empirical material below is generated automatically from the current repository outputs.

4 Reproducible June 2025 Greece–Italy Results

This section is generated automatically by the repository workflow from the empirical output files. The underlying inputs are benchmark yields transformed according to the repository's declared modeling convention; the resulting curves are therefore NSS model-implied discount factors rather than directly observed zero-coupon discount factors.

Table 1: NSS estimation diagnostics generated by the empirical pipeline.

Country	β_0	β_1	β_2	β_3	τ_1	τ_2	RMSE
Greece	0.35732110	-0.34046812	-0.14809965	-0.81407869	5.5000	25.5000	0.00004435
Italy	0.04604401	-0.02165477	-0.04818816	0.00020321	1.5000	8.0000	0.00000000

Table 2: NSS model-implied Greece–Italy discount-factor comparison.

Maturity	Greece	Italy	$\Delta D = D_{\mathrm{GR}} - D_{\mathrm{IT}}$
0.5	0.99121119	0.98946411	0.00174709
1.0	0.98164827	0.98037536	0.00127292
2.0	0.96006845	0.96041786	-0.00034941
3.0	0.93531750	0.93500818	0.00030932
5.0	0.87818206	0.87109869	0.00708337
7.0	0.81520694	0.80087065	0.01433629
10.0	0.72014715	0.69977250	0.02037466
15.0	0.58085573	0.55608047	0.02477526
20.0	0.46969468	0.44163295	0.02806173
30.0	0.29044017	0.27859393	0.01184624

5 Reproducibility and Interpretation

The pipeline succeeds only when empirical estimation, visualization, and the complete unit and regression suite succeed before paper publication. The resulting PDF therefore remains conditional on the tracked data, the declared transformation assumption, and the NSS implementation tested by continuous integration.

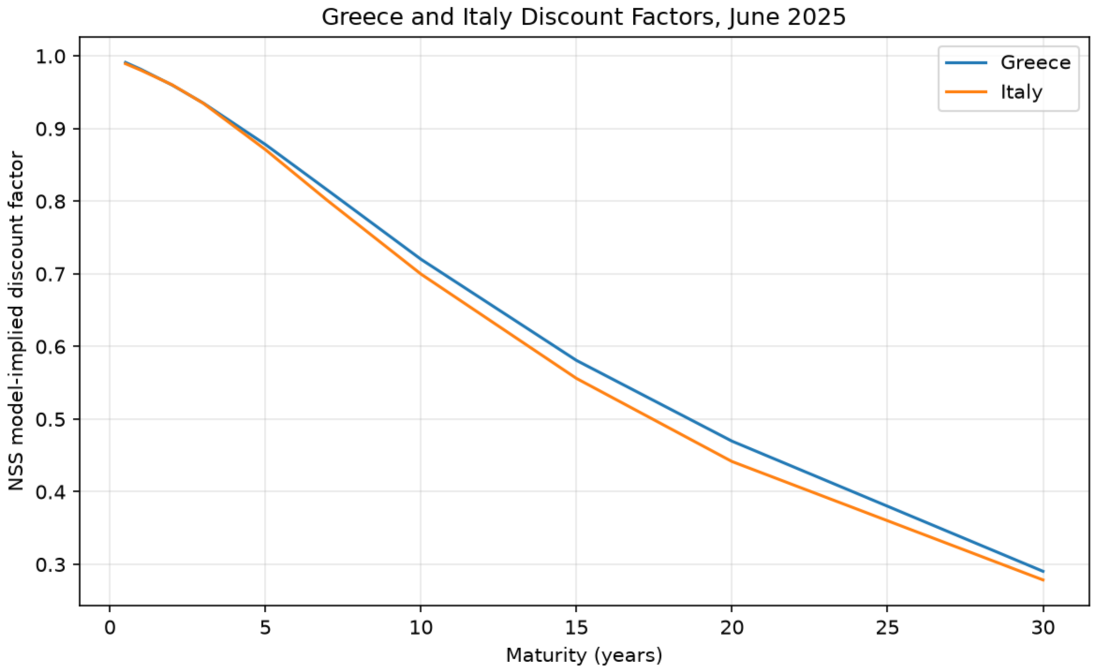

Figure 1: Automatically generated NSS model-implied discount-factor comparison.

References

[1] Nelson, C. R. and Siegel, A. F. (1987). Parsimonious modeling of yield curves. *Journal of Business*, 60(4), 473–489.

[2] Svensson, L. E. O. (1994). Estimating and interpreting forward interest rates: Sweden 1992–1994. *NBER Working Paper*.

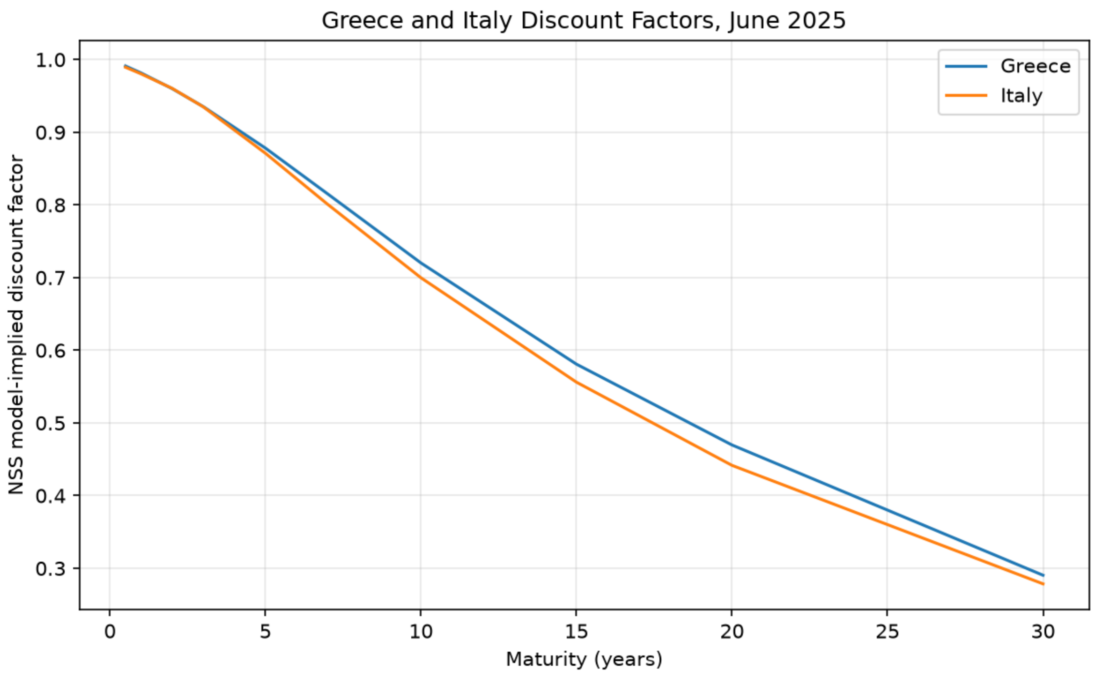

Figure 2: Greece and Italy NSS model-implied discount factors for the repository's June 2025 comparison period.